

ECON484 project

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Predicting Employment Outcomes among Young Adults: Education, Well-being, and Machine Learning Evidence from PSID-TAS 2019–2023.

Main difference between this and the original: added K-fold CV functions for CART, RF, and LASSO. This allows us to compare predictive performance of well-being vs education models more robustly. And, I implemented some modifications to apply them to panel data. Moreover, I used lagged well-being indicators to predict current employment, which is more realistic and avoids simultaneity bias.

```
library(dplyr)
library(ggplot2)
library(glmnet)
library(randomForest)
library(rpart)
library(rpart.plot)
library(lmtest)      # coeftest()
library(sandwich)    # vcovCL() for clustered SEs

panel <- read.csv("data/J362431/psid_tas_panel.csv")

panel <- panel %>%
  filter(emp1 != 99) %>%
  mutate(
    employed = ifelse(emp1 == 1, 1, 0),
    year      = factor(year)
  )

cat("Person-waves:", nrow(panel),
    "| Persons:", n_distinct(panel$person_id),
    "| Waves:", paste(levels(panel$year), collapse = "/"), "\n")

## Person-waves: 7281 | Persons: 3899 | Waves: 2019/2021/2023
table(panel$employed, panel$year)

##
##      2019 2021 2023
## 0 1143 1025 1080
## 1 1422 1290 1321
```

```

# -----
# K-fold cross-validation helpers
# Folds are assigned at the PERSON level (grouped CV): all observations
# for a given person go into the same fold, preventing data leakage
# across waves of the same individual.
# -----

# Assign each unique person to a fold; return fold index per row.
make_folds <- function(person_ids, k, seed = 123) {
  set.seed(seed)
  persons <- unique(person_ids)
  person_fold <- setNames(sample(rep_len(seq_len(k), length(persons))),
                           persons)
  unname(person_fold[as.character(person_ids)])
}

# K-fold CV accuracy for CART (rpart).
cv_cart <- function(data, person_ids, k, seed = 123) {
  folds <- make_folds(person_ids, k, seed)
  accs <- sapply(seq_len(k), function(f) {
    train <- data[folds != f, ]
    test <- data[folds == f, ]
    m <- rpart(employed ~ ., data = train, method = "class")
    pred <- as.integer(as.character(predict(m, test, type = "class")))
    mean(pred == test$employed)
  })
  mean(accs)
}

# K-fold CV accuracy for Random Forest.
cv_rf <- function(data, person_ids, k, seed = 123) {
  folds <- make_folds(person_ids, k, seed)
  accs <- sapply(seq_len(k), function(f) {
    set.seed(seed + f) # reproducible per fold
    train <- data[folds != f, ]
    test <- data[folds == f, ]
    m <- randomForest(as.factor(employed) ~ ., data = train, ntree = 300)
    pred <- as.integer(as.character(predict(m, test)))
    mean(pred == test$employed)
  })
  mean(accs)
}

# K-fold CV accuracy for LASSO.
# Outer loop: k person-grouped folds (same split as CART/RF for fair comparison).
# Inner loop: cv.glmnet on the training fold selects lambda (default 10-fold).
# This avoids the cv.glmnet nfolds >= 3 restriction when k = 2.
cv_lasso <- function(x, y, person_ids, k, seed = 123) {
  folds <- make_folds(person_ids, k, seed)
  accs <- sapply(seq_len(k), function(f) {
    set.seed(seed + f)
    x_tr <- x[folds != f, , drop = FALSE]; y_tr <- y[folds != f]
    x_te <- x[folds == f, , drop = FALSE]; y_te <- y[folds == f]
  })
  mean(accs)
}

```

```

fit <- cv.glmnet(x_tr, y_tr, family = "binomial", alpha = 1)
prob <- predict(fit, newx = x_te, s = "lambda.1se", type = "response")
mean(as.integer(prob > 0.5) == y_te)
})
mean(accs)
}

K_VALS <- c(2, 5, 10)

```

Well-being

```

wb_vars <- c("wb_happiness_lag", "wb_interest_lag", "wb_satisfied_lag",
            "wb_belong_lag", "wb_society_better_lag", "wb_people_good_lag",
            "wb_manage_daily_lag", "wb_trusting_rel_lag",
            "wb_confident_ideas_lag", "wb_life_direction_lag")

panel <- panel %>%
  mutate(across(all_of(wb_vars), ~ifelse(. %in% c(8, 9), NA, .)))

panel <- panel %>%
  rowwise() %>%
  mutate(wellbeing_score = mean(c_across(all_of(wb_vars)), na.rm = TRUE)) %>%
  ungroup()

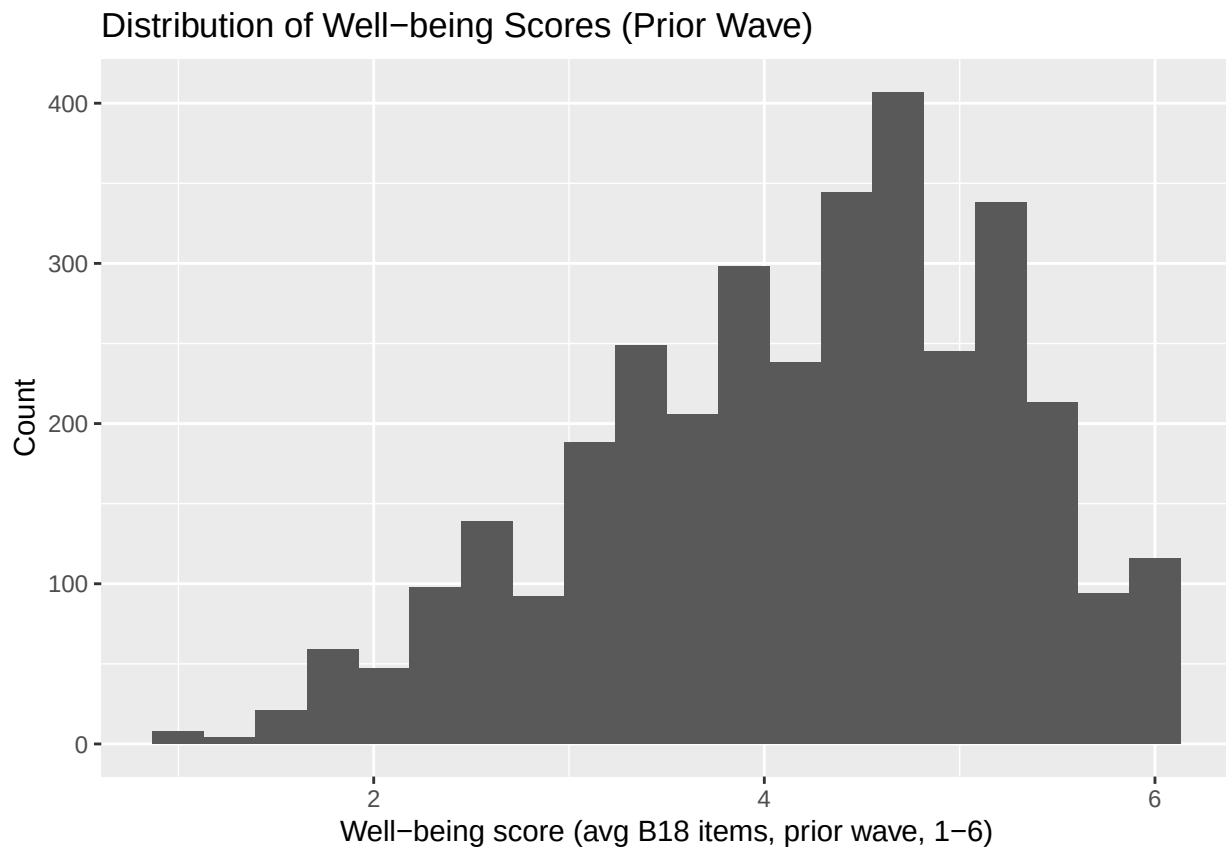
summary(panel$wellbeing_score)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##      1.00   3.50   4.30   4.18   5.00   6.00   3877

ggplot(panel, aes(x = wellbeing_score)) +
  geom_histogram(bins = 20) +
  labs(title = "Distribution of Well-being Scores (Prior Wave)",
       x = "Well-being score (avg B18 items, prior wave, 1-6)", y = "Count")

## Warning: Removed 3877 rows containing non-finite outside the scale range
## (`stat_bin()`).

```



```

panel %>%
  group_by(employed) %>%
  summarise(avg_wellbeing = mean(wellbeing_score, na.rm = TRUE))

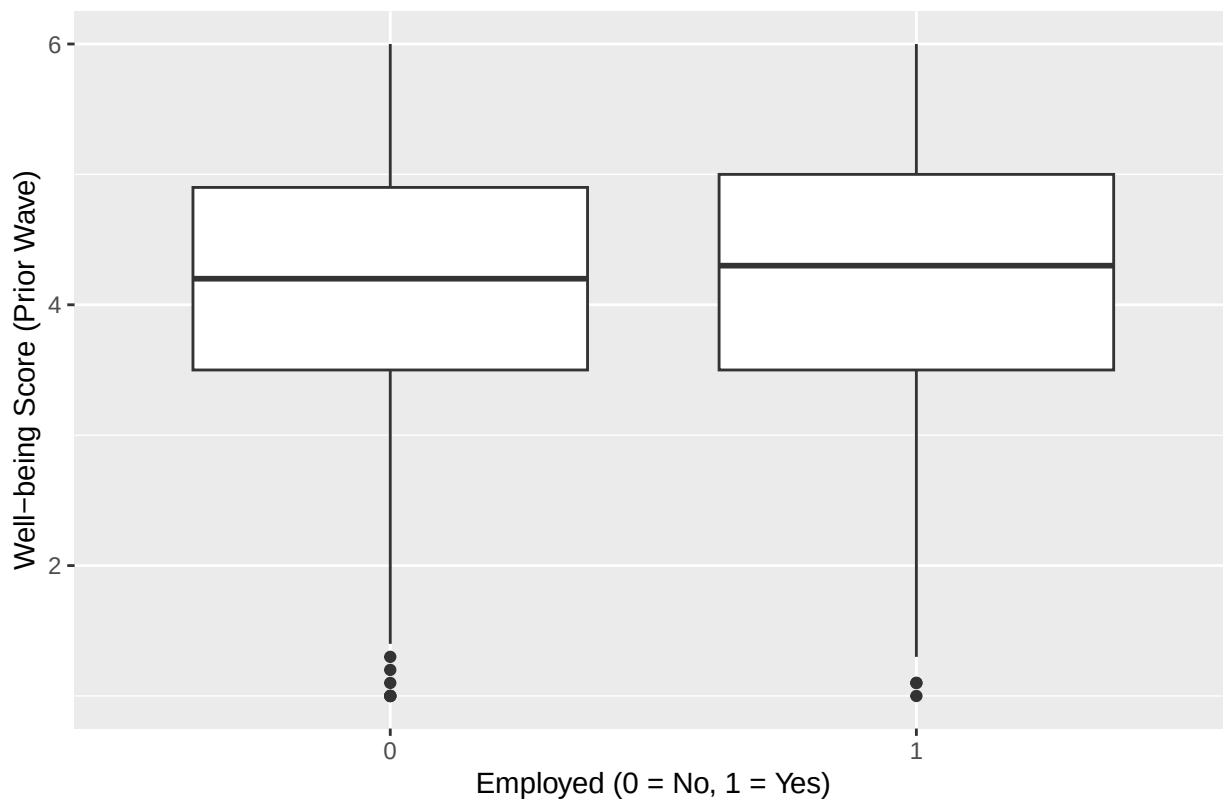
## # A tibble: 2 x 2
##   employed avg_wellbeing
##   <dbl>     <dbl>
## 1      0         4.15
## 2      1         4.20

ggplot(panel, aes(x = factor(employed), y = wellbeing_score)) +
  geom_boxplot() +
  labs(x = "Employed (0 = No, 1 = Yes)", y = "Well-being Score (Prior Wave)",
       title = "Prior-Wave Well-being by Employment Status")

## Warning: Removed 3877 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

```

Prior-Wave Well-being by Employment Status



```
wellbeing_logit <- glm(
  employed ~ wellbeing_score + year,
  data = panel, family = binomial
)
coeftest(wellbeing_logit,
  vcov = vcovCL(wellbeing_logit, cluster = ~person_id))
```

```
##
## z test of coefficients:
##
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.2973995  0.1639038  1.8145  0.0696 .
## wellbeing_score 0.0456602  0.0361789  1.2621  0.2069
## year2023        0.0067947  0.0647913  0.1049  0.9165
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Higher well-being have no association with employment status in the subsequent wave.

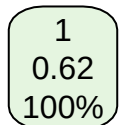
```
# Extract person_ids BEFORE renaming so they align with the data rows.
wb_complete <- panel %>%
  select(person_id, employed, year, all_of(wb_vars)) %>%
  na.omit()

wb_person_ids <- wb_complete$person_id # kept for grouped CV fold assignment

wb_tree_data <- wb_complete %>%
  select(-person_id)
```

```
colnames(wb_tree_data) <- c(
  "employed", "Year",
  "Happy", "InterestInLife", "Satisfied",
  "Community", "SocietyBetter", "PeopleGood",
  "ManageResponsibilities", "TrustingRelationships",
  "Confidence", "LifePurpose"
)
```

```
wellbeing_tree <- rpart(employed ~ ., data = wb_tree_data, method = "class")
rpart.plot(wellbeing_tree)
```



```
printcp(wellbeing_tree)
```

```
##
## Classification tree:
## rpart(formula = employed ~ ., data = wb_tree_data, method = "class")
##
## Variables actually used in tree construction:
## character(0)
##
## Root node error: 1272/3362 = 0.37835
##
## n= 3362
##
##   CP nsplit rel error xerror xstd
## 1  0      0         1      0     0
```

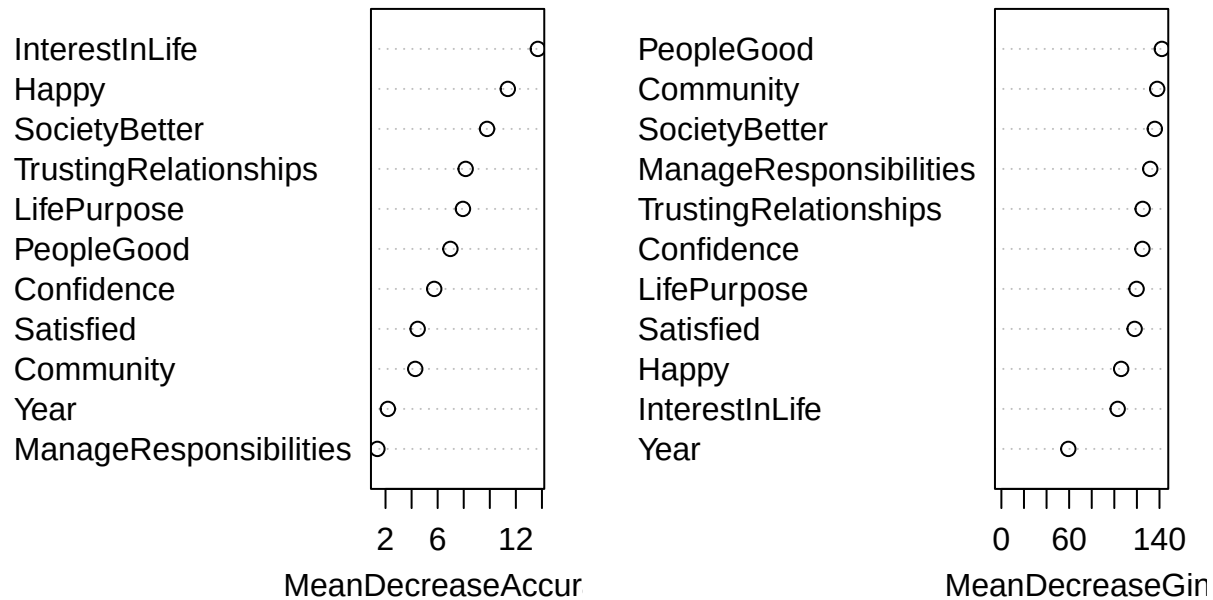
The CART model failed to identify meaningful splits, suggesting that individual well-being indicators alone do not strongly distinguish employed from non-employed young adults.

```
set.seed(123)
wellbeing_rf <- randomForest(
  as.factor(employed) ~ .,
  data = wb_tree_data, ntree = 500, importance = TRUE
)
wellbeing_rf
```

```
##
## Call:
## randomForest(formula = as.factor(employed) ~ ., data = wb_tree_data,      ntree = 500, importance =
##               Type of random forest: classification
##               Number of trees: 500
## No. of variables tried at each split: 3
##
##               OOB estimate of  error rate: 42.21%
## Confusion matrix:
##      0      1 class.error
## 0 227 1045  0.8215409
## 1 374 1716  0.1789474
```

```
varImpPlot(wellbeing_rf)
```

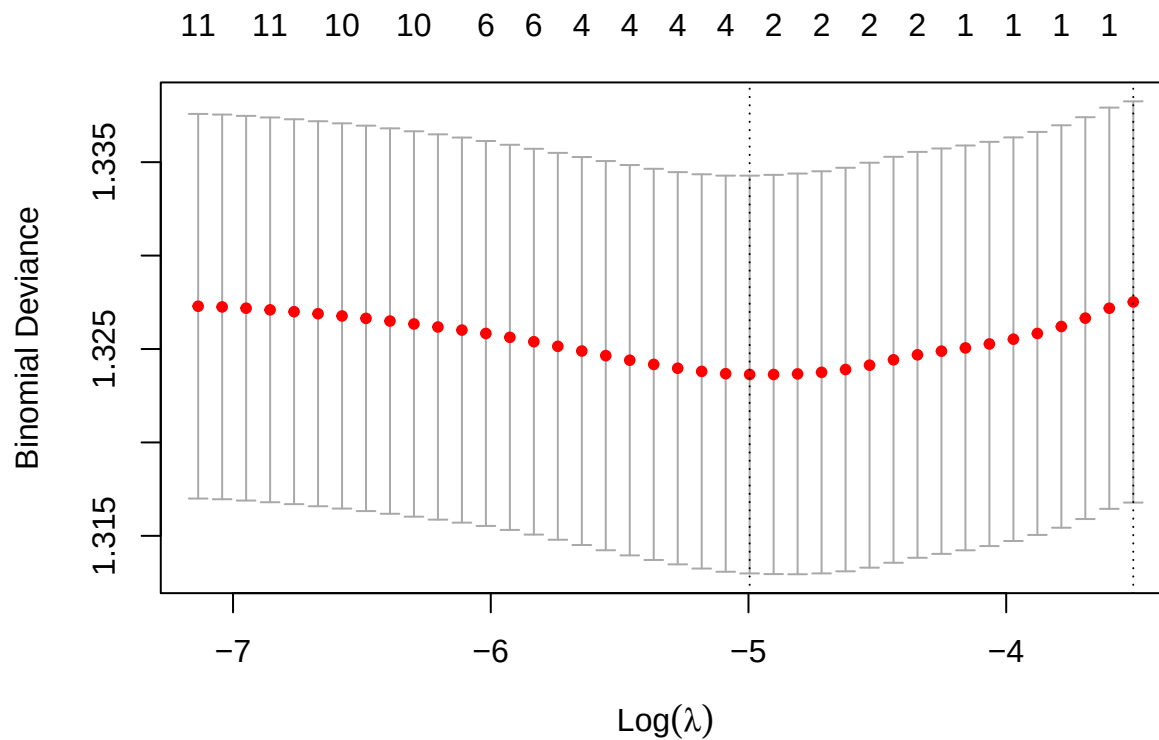
wellbeing_rf



Some well-being indicators appear moderately important, but predictive power remains weak.

```
x_wb <- model.matrix(employed ~ ., wb_tree_data)[, -1]
y_wb <- wb_tree_data$employed

set.seed(123)
wellbeing_lasso <- cv.glmnet(x_wb, y_wb, family = "binomial", alpha = 1)
plot(wellbeing_lasso)
```



```
coef(wellbeing_lasso, s = "lambda.min")
```

```
## 13 x 1 sparse Matrix of class "dgCMatrix"
##                               s1
## (Intercept)                   0.220807342
## Year2021                       .
## Year2023                       .
## Happy                         .
## InterestInLife                 .
## Satisfied                      .
## Community                     .
## SocietyBetter                  -0.028223262
## PeopleGood                    .
## ManageResponsibilities         .
## TrustingRelationships          0.079809083
## Confidence                     .
## LifePurpose                    -0.002015184
```

```
coef(wellbeing_lasso, s = "lambda.1se")
```

```
## 13 x 1 sparse Matrix of class "dgCMatrix"
##                               s1
## (Intercept)                   0.4965736
## Year2021                       .
## Year2023                       .
## Happy                         .
## InterestInLife                 .
## Satisfied                      .
## Community                     .
## SocietyBetter                  .
## PeopleGood                    .
```

```
## ManageResponsibilities .
## TrustingRelationships .
## Confidence .
## LifePurpose .
```

lambda.1se selected no predictors.

```
# K-fold CV (grouped by person) for k = 2, 5, 10.
# Folds are split at the person level to avoid leakage across waves.
wb_cv <- data.frame(
  k = K_VALS,
  CART = sapply(K_VALS, cv_cart, data = wb_tree_data, person_ids = wb_person_ids),
  RF = sapply(K_VALS, cv_rf, data = wb_tree_data, person_ids = wb_person_ids),
  LASSO = sapply(K_VALS, cv_lasso, x = x_wb, y = y_wb, person_ids = wb_person_ids)
)

cat("Well-being model CV accuracy\n")
```

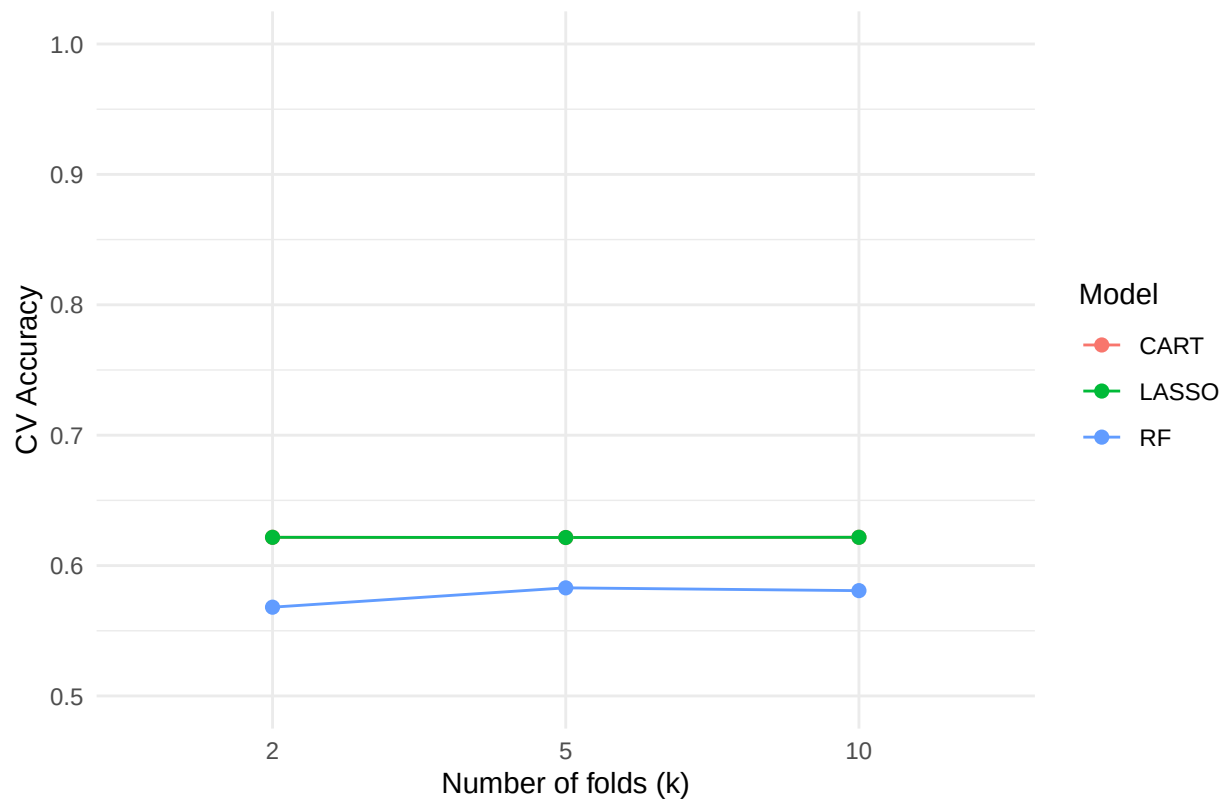
```
## Well-being model CV accuracy
```

```
print(round(wb_cv, 3))
```

```
##      k  CART    RF LASSO
## 1   2 0.622 0.568 0.622
## 2   5 0.622 0.583 0.622
## 3  10 0.622 0.581 0.622
```

```
# Visualise
wb_cv_long <- tidyr::pivot_longer(wb_cv, -k,
                                   names_to = "Model", values_to = "Accuracy")
ggplot(wb_cv_long, aes(x = factor(k), y = Accuracy,
                        colour = Model, group = Model)) +
  geom_line() + geom_point(size = 2) +
  scale_y_continuous(limits = c(0.5, 1)) +
  labs(title = "Well-being Models: CV Accuracy by k",
       x = "Number of folds (k)", y = "CV Accuracy") +
  theme_minimal()
```

Well-being Models: CV Accuracy by k



Education

```
edu_vars <- c("highest_edu", "grade_level", "attended_college",
             "edu_aspiration", "edu_expectation",
             "mother_edu", "father_edu")

panel <- panel %>%
  mutate(
    highest_edu      = ifelse(highest_edu == 99, NA, highest_edu),
    grade_level      = ifelse(grade_level %in% c(0, 99), NA, grade_level),
    attended_college = ifelse(attended_college == 9, NA, attended_college),
    edu_aspiration    = ifelse(edu_aspiration %in% c(8, 98, 99), NA,
                               edu_aspiration),
    edu_expectation  = ifelse(edu_expectation %in% c(0, 9), NA,
                               edu_expectation),
    mother_edu       = ifelse(mother_edu %in% c(0, 96, 99), NA, mother_edu),
    father_edu       = ifelse(father_edu %in% c(0, 96, 98, 99), NA, father_edu)
  )

edu_logit_data <- panel %>%
  select(person_id, employed, year, all_of(edu_vars)) %>%
  na.omit()

cat("Education logit sample:", nrow(edu_logit_data), "person-waves\n")
```

```

## Education logit sample: 190 person-waves
education_logit <- glm(
  employed ~ . - person_id,
  data = edu_logit_data, family = binomial
)
coeftest(education_logit,
  vcov = vcovCL(education_logit, cluster = ~person_id))

##
## z test of coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.648205   2.118665  0.3059 0.759643
## year2021       0.935986   0.407735  2.2956 0.021700 *
## year2023       0.481629   0.427371  1.1270 0.259760
## highest_edu    0.455863   0.114236  3.9905 6.593e-05 ***
## grade_level   -0.189654   0.135100 -1.4038 0.160376
## attended_college 0.440415  0.169050  2.6052 0.009181 **
## edu_aspiration  0.081648   0.316017  0.2584 0.796126
## edu_expectation -0.061029  0.302484 -0.2018 0.840106
## mother_edu     -0.040511  0.084585 -0.4789 0.631982
## father_edu     -0.075115  0.088026 -0.8533 0.393474
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# Extract person_ids BEFORE renaming.
edu_complete <- panel %>%
  select(person_id, employed, year, all_of(edu_vars)) %>%
  na.omit()

edu_person_ids <- edu_complete$person_id

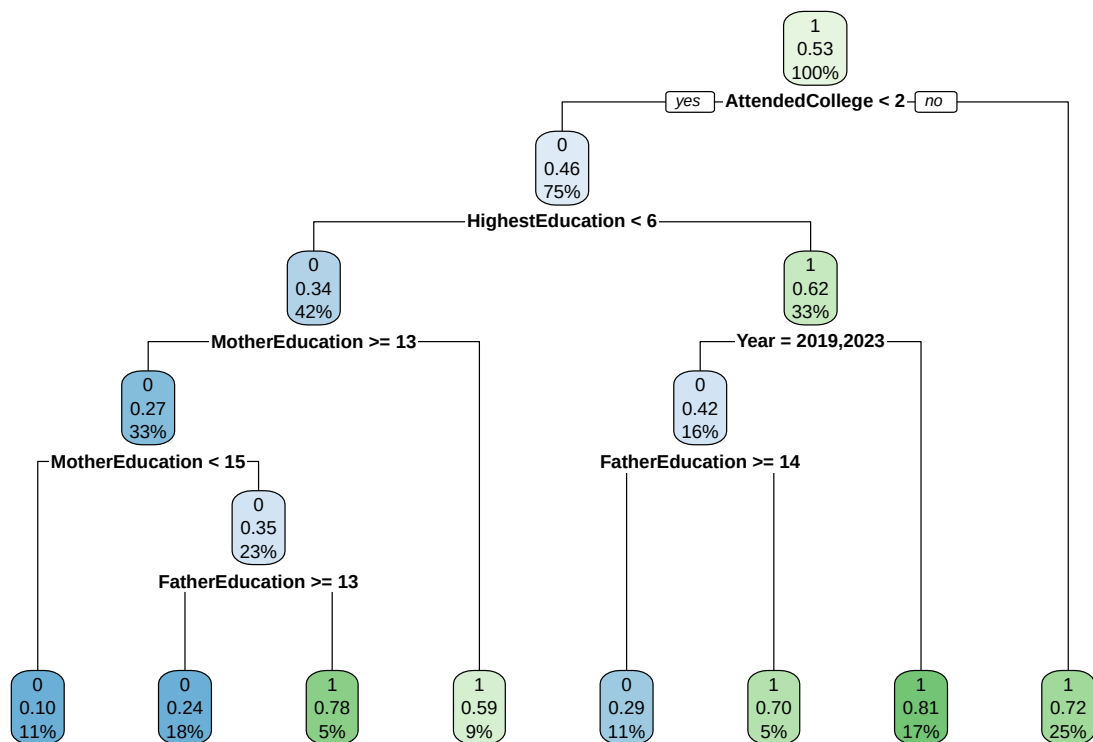
edu_tree_data <- edu_complete %>%
  select(-person_id)

colnames(edu_tree_data) <- c(
  "employed", "Year",
  "HighestEducation", "GradeCompleted",
  "AttendedCollege",
  "EducationAspiration", "EducationExpectation",
  "MotherEducation", "FatherEducation"
)

cat("Education ML sample:", nrow(edu_tree_data), "person-waves\n")

## Education ML sample: 190 person-waves
education_tree <- rpart(employed ~ ., data = edu_tree_data, method = "class")
rpart.plot(education_tree)

```

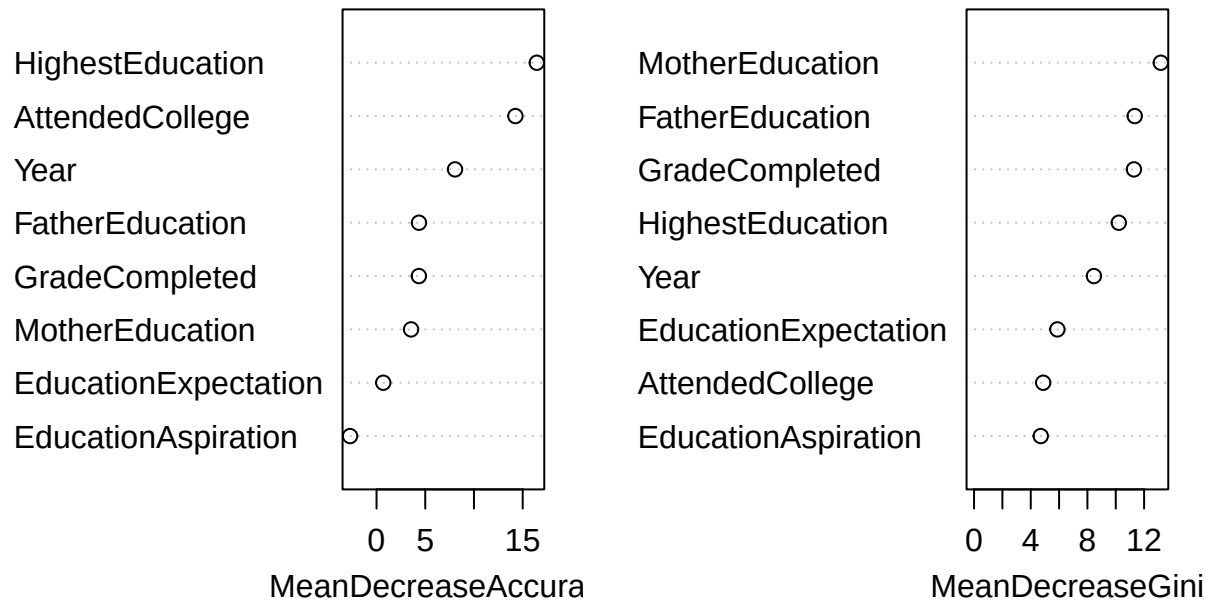


```

set.seed(123)
education_rf <- randomForest(
  as.factor(employed) ~ .,
  data = edu_tree_data, ntree = 500, importance = TRUE
)
varImpPlot(education_rf)

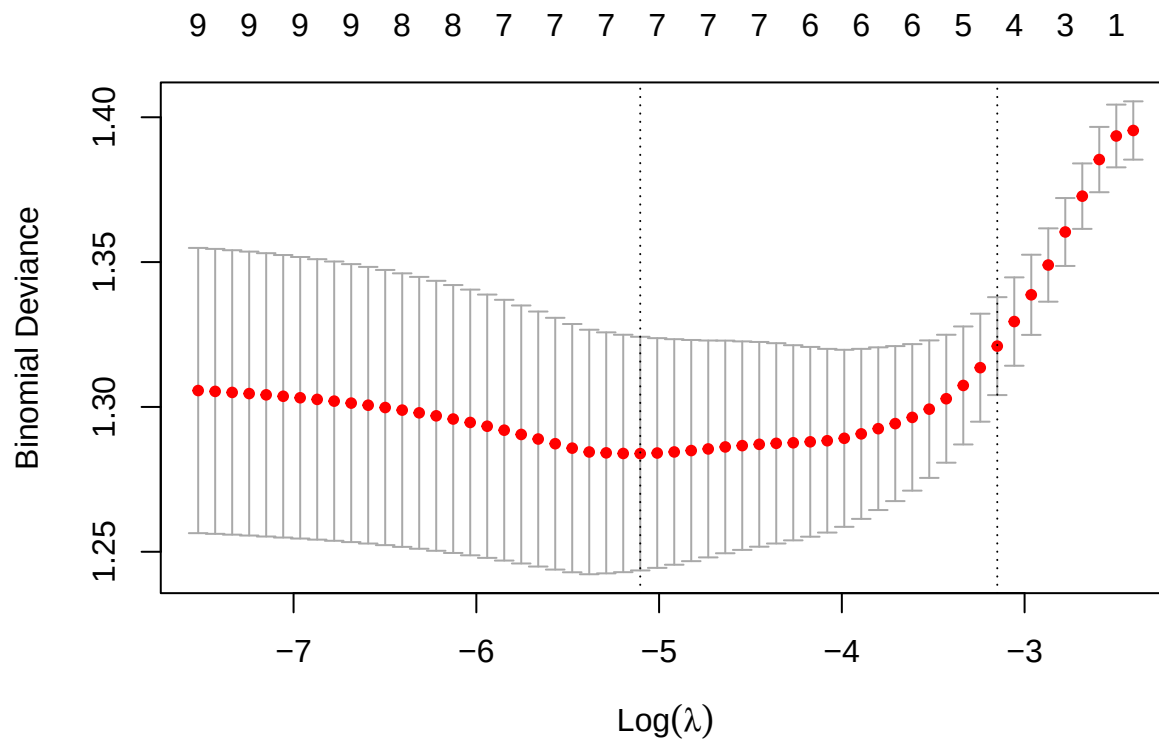
```

education_rf



```
x_edu <- model.matrix(employed ~ ., edu_tree_data)[, -1]
y_edu <- edu_tree_data$employed

set.seed(123)
education_lasso <- cv.glmnet(x_edu, y_edu, family = "binomial", alpha = 1)
plot(education_lasso)
```



```
coef(education_lasso, s = "lambda.min")
```

```
## 10 x 1 sparse Matrix of class "dgCMatrix"
##                               s1
## (Intercept)          0.47841292
## Year2021             0.74375330
## Year2023             0.27235934
## HighestEducation     0.39246668
## GradeCompleted      -0.14580621
## AttendedCollege      0.39402017
## EducationAspiration  .
## EducationExpectation .
## MotherEducation     -0.03706409
## FatherEducation     -0.06609582
```

```
coef(education_lasso, s = "lambda.1se")
```

```
## 10 x 1 sparse Matrix of class "dgCMatrix"
##                               s1
## (Intercept)          -0.8804412869
## Year2021             0.2113264375
## Year2023             .
## HighestEducation     0.1369513183
## GradeCompleted       .
## AttendedCollege      0.2440483812
## EducationAspiration  .
## EducationExpectation .
## MotherEducation     -0.0009203347
## FatherEducation     -0.0260473095
```

```
edu_cv <- data.frame(
  k      = K_VALS,
```

```

CART = sapply(K_VALS, cv_cart, data = edu_tree_data, person_ids = edu_person_ids),
RF    = sapply(K_VALS, cv_rf,   data = edu_tree_data, person_ids = edu_person_ids),
LASSO = sapply(K_VALS, cv_lasso, x = x_edu, y = y_edu, person_ids = edu_person_ids)
)

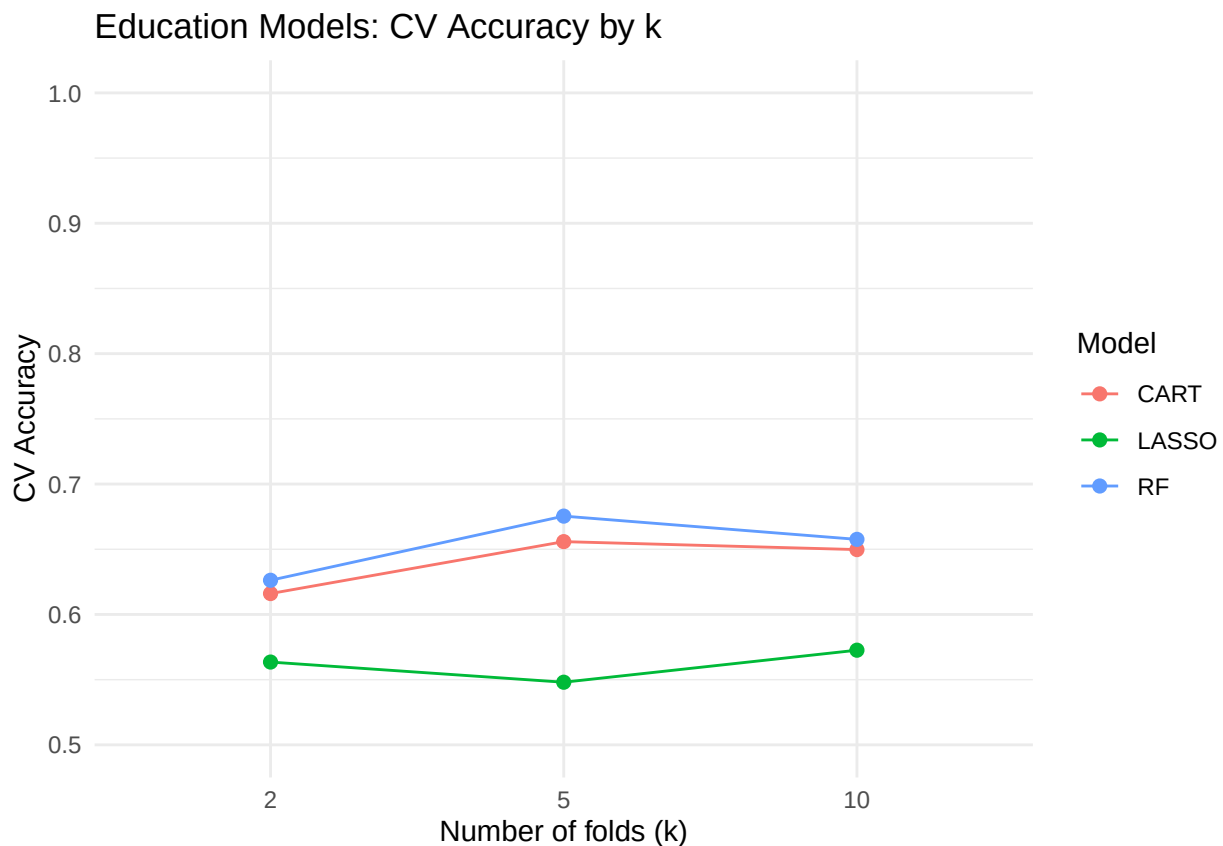
cat("Education model CV accuracy\n")

## Education model CV accuracy
print(round(edu_cv, 3))

##   k  CART    RF LASSO
## 1  2 0.616 0.626 0.563
## 2  5 0.656 0.675 0.548
## 3 10 0.650 0.658 0.573

edu_cv_long <- tidyr::pivot_longer(edu_cv, ~k,
                                   names_to = "Model", values_to = "Accuracy")
ggplot(edu_cv_long, aes(x = factor(k), y = Accuracy,
                       colour = Model, group = Model)) +
  geom_line() + geom_point(size = 2) +
  scale_y_continuous(limits = c(0.5, 1)) +
  labs(title = "Education Models: CV Accuracy by k",
       x = "Number of folds (k)", y = "CV Accuracy") +
  theme_minimal()

```



Comparison: Well-being vs Education

```

wellbeing_AIC <- AIC(wellbeing_logit)
education_AIC <- AIC(education_logit)

wb_tree_splits <- wellbeing_tree$cptable[nrow(wellbeing_tree$cptable), "nsplit"]
edu_tree_splits <- education_tree$cptable[nrow(education_tree$cptable), "nsplit"]

wb_lasso_vars <- sum(coef(wellbeing_lasso, s = "lambda.1se")[-1, ] != 0)
edu_lasso_vars <- sum(coef(education_lasso, s = "lambda.1se")[-1, ] != 0)

wb_rf_acc <- mean(predict(wellbeing_rf, wb_tree_data) == wb_tree_data$employed)
edu_rf_acc <- mean(predict(education_rf, edu_tree_data) == edu_tree_data$employed)

comparison_table <- data.frame(
  Model_Group      = c("Well-being", "Education"),
  Logit_AIC        = c(wellbeing_AIC, education_AIC),
  Tree_Splits      = c(wb_tree_splits, edu_tree_splits),
  LASSO_1se_Vars   = c(wb_lasso_vars, edu_lasso_vars),
  RF_Training_Accuracy = c(wb_rf_acc, edu_rf_acc)
)

comparison_table

```

```

##   Model_Group Logit_AIC Tree_Splits LASSO_1se_Vars RF_Training_Accuracy
## 1 Well-being 4523.6508          0          0          0.9589530
## 2 Education 248.3171          7          5          0.9210526

# Side-by-side CV accuracy table: well-being vs education, for k = 2, 5, 10.
cv_summary <- data.frame(
  k      = K_VALS,
  WB_CART = wb_cv$CART,
  WB_RF   = wb_cv$RF,
  WB_LASSO = wb_cv$LASSO,
  Edu_CART = edu_cv$CART,
  Edu_RF   = edu_cv$RF,
  Edu_LASSO = edu_cv$LASSO
)

cat("CV accuracy: well-being vs education models\n")

```

```
## CV accuracy: well-being vs education models
```

```
print(round(cv_summary, 3))
```

```

##      k WB_CART WB_RF WB_LASSO Edu_CART Edu_RF Edu_LASSO
## 1  2  0.622 0.568  0.622  0.616 0.626  0.563
## 2  5  0.622 0.583  0.622  0.656 0.675  0.548
## 3 10  0.622 0.581  0.622  0.650 0.658  0.573

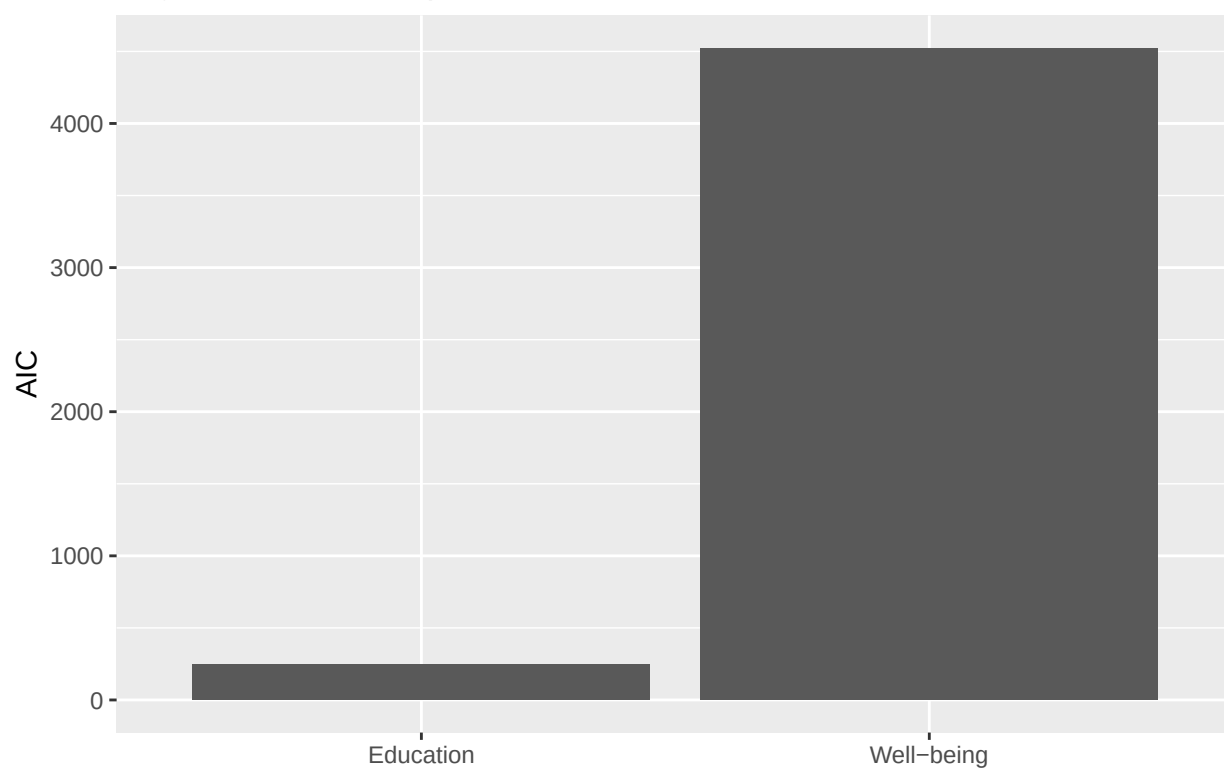
```

```

ggplot(comparison_table, aes(x = Model_Group, y = Logit_AIC)) +
  geom_col() +
  labs(title = "Logit Model Fit Comparison", x = "", y = "AIC")

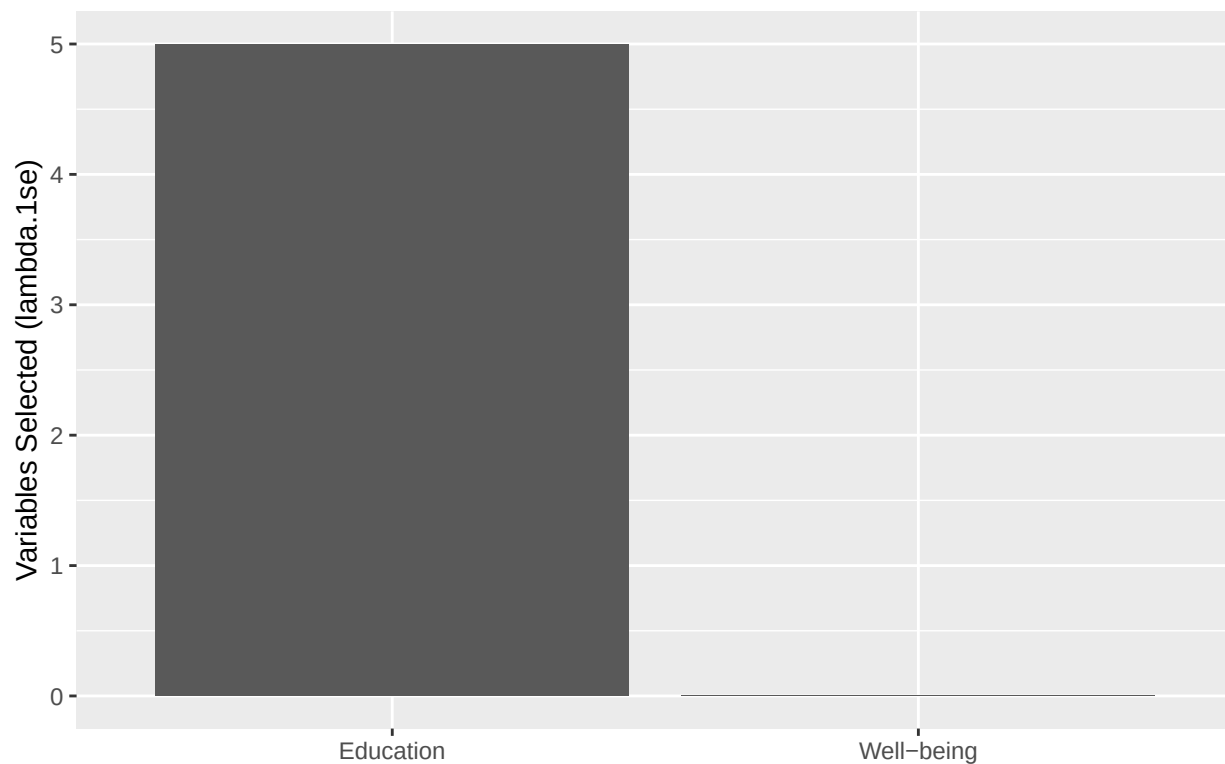
```

Logit Model Fit Comparison



```
ggplot(comparison_table, aes(x = Model_Group, y = LASSO_1se_Vars)) +  
  geom_col() +  
  labs(title = "LASSO Variable Selection Comparison",  
        x = "", y = "Variables Selected (lambda.1se)")
```

LASSO Variable Selection Comparison



Combined Model: Does Well-being Add Anything to Education?

```
combined_data <- panel %>%
  select(person_id, employed, year, wellbeing_score, all_of(edu_vars)) %>%
  na.omit()

colnames(combined_data) <- c(
  "person_id", "employed", "Year", "WellbeingScore",
  "HighestEducation", "GradeCompleted", "AttendedCollege",
  "EducationAspiration", "EducationExpectation",
  "MotherEducation", "FatherEducation"
)

cat("Combined-model sample size:", nrow(combined_data), "person-waves\n")

## Combined-model sample size: 120 person-waves

combined_edu_only <- glm(
  employed ~ . - WellbeingScore - person_id,
  data = combined_data, family = binomial
)

combined_full <- glm(
  employed ~ . - person_id,
  data = combined_data, family = binomial
)

coefTest(combined_full,
```

```

vcov = vcovCL(combined_full, cluster = ~person_id))

##
## z test of coefficients:
##
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)      3.8351099  2.7708621  1.3841  0.16633
## Year2023         -0.4002647  0.4030443 -0.9931  0.32066
## WellbeingScore   -0.1173301  0.1919041 -0.6114  0.54094
## HighestEducation  0.44400202  0.1391006  3.1633  0.00156 **
## GradeCompleted   -0.2459954  0.1729040 -1.4227  0.15481
## AttendedCollege   0.2260445  0.2325665  0.9720  0.33107
## EducationAspiration 0.3876458  0.4351585  0.8908  0.37303
## EducationExpectation -0.3281833  0.4056188 -0.8091  0.41846
## MotherEducation   -0.1428831  0.1081711 -1.3209  0.18654
## FatherEducation   -0.0070526  0.1030715 -0.0684  0.94545
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

anova(combined_edu_only, combined_full, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: employed ~ (person_id + Year + WellbeingScore + HighestEducation +
##   GradeCompleted + AttendedCollege + EducationAspiration +
##   EducationExpectation + MotherEducation + FatherEducation) -
##   WellbeingScore - person_id
## Model 2: employed ~ (person_id + Year + WellbeingScore + HighestEducation +
##   GradeCompleted + AttendedCollege + EducationAspiration +
##   EducationExpectation + MotherEducation + FatherEducation) -
##   person_id
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      111      145.04
## 2      110      144.64  1  0.40078  0.5267

AIC(combined_edu_only, combined_full)

##               df      AIC
## combined_edu_only  9 163.0415
## combined_full      10 164.6407

```